Assignment One :

Practical Python EDA and Decision Tree Case Study

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DATA INSPECTION, CLEANING AND MODEL RESULTS

```
--- 1. DATASET OVERVIEW ---
Shape: (10, 8)

Data types:
 Student_ID              int64
County                   str
Study_Hours_Per_Week    float64
Attendance_Percent      float64
Fee_Balance_Ksh           int64
Previous_Grade            int64
Student_Type                str
Academic_Risk             int64
dtype: object

Summary statistics:
        Student_ID  County  Study_Hours_Per_Week  Attendance_Percent  Fee_Balance_Ksh  Previous_Grade Student_Type  Academic_Risk
count    10.000000      9              9.000000            9.000000        10.000000       10.000000           10      10.000000
unique         NaN      4                   NaN                 NaN              NaN             NaN            2             NaN
top            NaN  Nakuru                   NaN                 NaN              NaN             NaN      Regular             NaN
freq           NaN      4                   NaN                 NaN              NaN             NaN            7             NaN
mean   1004.600000    NaN             17.000000           78.111111      7700.000000       66.700000          NaN       0.400000
std       2.875181    NaN              7.123903           14.598326      9153.627089       16.145519          NaN       0.516398
min    1001.000000    NaN              8.000000           55.000000         0.000000       45.000000          NaN       0.000000
25%    1002.250000    NaN             12.000000           65.000000         0.000000       52.000000          NaN       0.000000
50%    1004.500000    NaN             15.000000           85.000000      3500.000000       68.500000          NaN       0.000000
75%    1006.750000    NaN             20.000000           88.000000     14250.000000       79.500000          NaN       1.000000
max    1009.000000    NaN             30.000000           95.000000     25000.000000       90.000000          NaN       1.000000
```

This screenshot shows the initial inspection of the student academic risk dataset. It displays the dataset structure, number of records and columns, data types, and summary statistics for the available variables.

screenshot 1 :Data set overview

```
--- 2. DATA QUALITY ---
Missing values:
 Student_ID                0
County                     1
Study_Hours_Per_Week       1
Attendance_Percent         1
Fee_Balance_Ksh            0
Previous_Grade             0
Student_Type               0
Academic_Risk              0
dtype: int64
Duplicate rows: 1
Duplicate Student_IDs: 1
```

This screenshot presents the data quality checks performed on the dataset. The results identify missing values in County, Study_Hours_Per_Week, and Attendance_Percent, as well as duplicate rows and duplicate Student_IDs.

screenshot 2 :Data Quality

```
--- 3. CLEANED DATA ---
     County  Study_Hours_Per_Week  Attendance_Percent  Fee_Balance_Ksh  Previous_Grade Student_Type  Academic_Risk Attendance_Category  Has_Fee_Balance
0    Nakuru                  15.0                85.0                0              72      Regular              0              High                0
1     Narok                   8.0                60.0            15000              48      Regular              1            Medium                1
2    Nakuru                  20.0                92.0                0              85  Scholarship              0              High                0
3     Narok                  15.0                55.0            25000              45      Regular              1               Low                1
4  KaKamega                  12.0                78.0             5000              65  Scholarship              0            Medium                1
6   Nairobi                  30.0                95.0                0              90      Regular              0              High                0
7    Nakuru                  10.0                65.0            18000              50  Scholarship              1            Medium                1
8     Narok                  25.0                88.0             2000              82      Regular              0              High                1
9    Nakuru                  18.0                85.0            12000              58      Regular              1              High                1
```

screenshot 3 : Clean Data

This screenshot displays the dataset after data cleaning and preprocessing. Missing values and duplicate records have been addressed, and additional features, including Attendance_Category and Has_Fee_Balance, have been created to support further analysis.

```
--- 4. DECISION TREE RESULTS ---
Training accuracy: 1.0

Classification report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00         5
           1       1.00      1.00      1.00         4

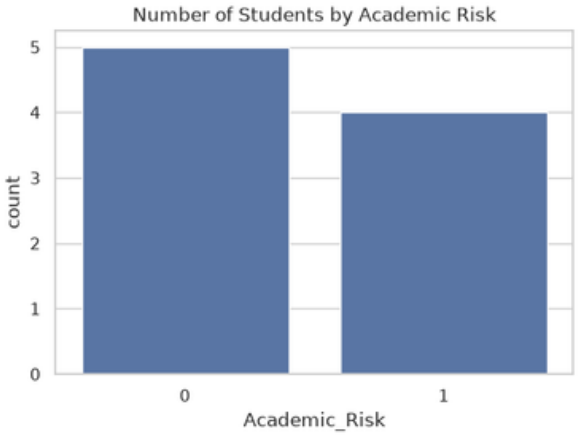
    accuracy                           1.00         9
   macro avg       1.00      1.00      1.00         9
weighted avg       1.00      1.00      1.00         9
```

scrreenshot 4 :Decision Tree Result.

This screenshot shows the performance of the Decision Tree classification model. The model achieved 100% training accuracy on the available dataset. However, because the dataset is small, further testing on unseen data is necessary to assess its ability to predict academic risk reliably.

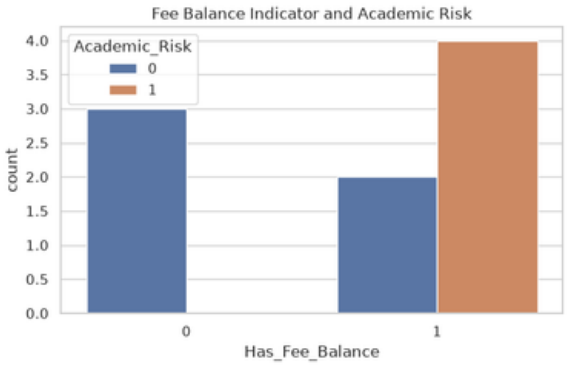
EXPLORATORY DATA ANALYSIS (EDA)

- **Exploratory Data Analysis (EDA) was performed to identify patterns and relationships in the student academic risk dataset. Visualizations were used to examine academic risk distribution, fee balance, attendance, study hours, and correlations between numerical variables.**



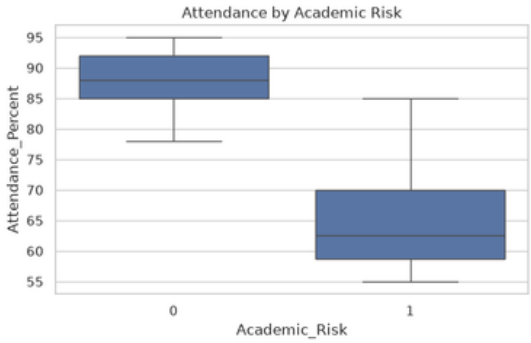
- **Figure 1: Distribution of Academic Risk**

This chart shows the distribution of students across the academic risk categories in the dataset. It helps identify which risk category contains the largest number of students.



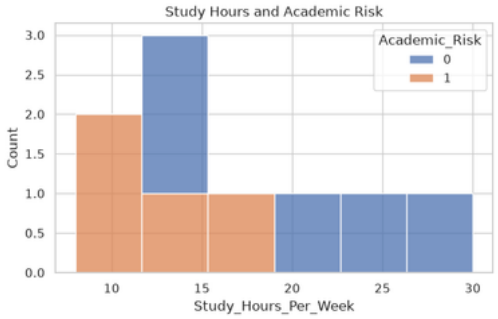
- **Figure 2: Fee Balance and Academic Risk**

This chart examines the relationship between students' fee balances and their academic risk categories.



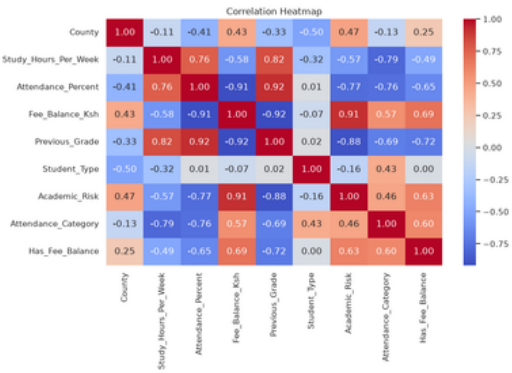
- **Figure 3: Attendance and Academic Risk**

This chart examines how attendance levels relate to the academic risk categories of students.



- **Figure 4: Study Hours and Academic Risk**

This chart explores the relationship between weekly study hours and students' academic risk categories.



- **Figure 5: Study Hours and Academic Risk**

This heatmap displays correlations among numerical variables, helping identify possible relationships between the dataset's features.

DECISION TREE CLASSIFICATION MODEL

A Decision Tree classifier was used to analyse student academic risk. The model learns patterns from the dataset's features and uses them to classify students into academic risk categories. The results were examined using training accuracy and a classification report.

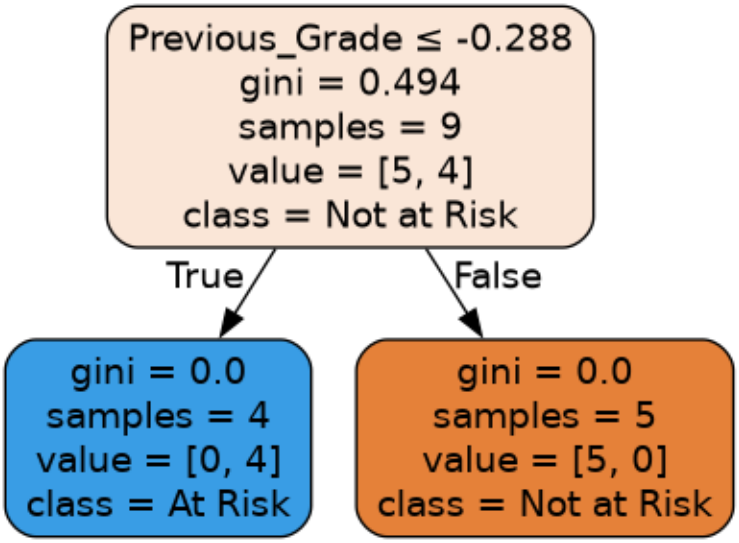


Figure 6: Decision Tree Model Visualization

This figure illustrates the structure of the trained Decision Tree classifier. The tree uses feature-based decision rules to divide the data into branches and produce academic risk predictions.

DISCUSSION OF FINDINGS

- The exploratory data analysis revealed several data quality issues, including missing values, duplicate records, and duplicate student identification numbers. These issues were addressed during data cleaning to improve the dataset's consistency.
- The visualizations were used to examine the distribution of academic risk and explore possible relationships between fee balance, attendance, and weekly study hours. The correlation heatmap also helped examine relationships among numerical variables.
- The Decision Tree classifier achieved 100% training accuracy on the available dataset. However, this result should be interpreted cautiously because the dataset is very small. The model may perform differently on unseen student records, so further evaluation using a separate test dataset or cross-validation is recommended.

LIMITATIONS OF THE STUDY

- Small dataset: The dataset contains only a few student records, which limits the reliability and generalizability of the findings.
- Data quality issues: Missing values and duplicate records required cleaning before analysis.
- Model evaluation: The reported accuracy is based on training data and does not independently establish how well the model predicts unseen cases.
- Limited features: Additional relevant variables could improve the analysis of student academic risk.

CONCLUSION

- This study demonstrated the use of Python for data inspection, cleaning, exploratory data analysis, and Decision Tree classification. The analysis identified data quality issues and explored patterns in student academic risk. The Decision Tree achieved 100% training accuracy, but further testing with a larger dataset and unseen records is necessary before relying on the model for real-world academic risk prediction.